

VaaniNotes

Offline, multilingual lecture and meeting notes that run on the Snapdragon NPU.

1. Summary

VaaniNotes is a Windows application for Snapdragon-powered PCs that listens to a lecture or meeting in Hindi, English or nine other Indian languages and produces a timestamped transcript, a summary, key points, action items with deadlines, revision flashcards, and answers to questions about the recording. Speech recognition and note writing both run on the Hexagon NPU using models from Qualcomm AI Hub. No audio, text or telemetry leaves the device. (Vaani means speech.)

2. Problem

- **Multilingual speech.** Indian classrooms and offices switch between English, Hindi and regional languages within one session; English-only note tools fail.
- **Unreliable connectivity.** Cloud transcription is unavailable in hostels, on trains, on rural campuses and in field offices.
- **Privacy.** Lectures, patient consultations, client calls and internal meetings should not be uploaded to a third party to obtain a summary.

3. Solution and key features

- **Speech to text in 11 languages** with automatic language detection, or direct speech-to-English translation.
- **Structured notes** in any of 11 output languages: summary, key points, action items and deadlines, flip-card flashcards.
- **Ask the recording:** natural-language questions answered from the transcript with [mm:ss] citations.
- **Private by construction:** the local server binds to 127.0.0.1 and the UI is served with a strict Content-Security-Policy (default-src 'self'), so the page cannot contact any external host.
- **Always usable:** NPU → CPU fallback for speech; GenieX → other local LLM server → built-in extractive engine for notes.

4. Technical implementation on Snapdragon

- **Speech model.** Whisper-Small (241M parameters) from Qualcomm AI Hub in the Precompiled QAIRT ONNX format, fetched for the exact chipset (Snapdragon X Elite, X Plus or X2 Elite).
- **Runtime.** ONNX Runtime with the QNN execution provider on the HTP (NPU) backend in burst performance mode; both the 2.x plugin provider and the legacy built-in provider are supported.
- **Decoder.** A numpy implementation of the KV-cache greedy decoding loop written against the AI Hub I/O contract (float16 named tensors: cross-attention cache from the encoder, self-attention cache carried between steps, forced language/task prompt). The NPU path requires no PyTorch, keeping the ARM64 install small.

- **Notes LLM.** Qwen3-4B-Instruct-2507 (w4a16, QAIRT) served on the NPU by Qualcomm GenieX through its local OpenAI-compatible endpoint. Long recordings are condensed map-reduce style to fit the 4K context.
- **Application.** FastAPI backend with a single-worker job queue that streams partial results; accessible vanilla HTML/CSS/JS frontend with self-hosted fonts; command-line interface; Markdown export.

5. Performance

Figures published by Qualcomm AI Hub (models v0.63.0) for the exact models used, measured on the NPU of hosted reference devices:

Model / component	Snapdragon X Elite	Snapdragon X2 Elite
Whisper-Small encoder (per 30 s of audio)	117 ms	53 ms
Whisper-Small decoder (per token)	10.5 ms	6.3 ms
Qwen3-4B-Instruct, GenieX QAIRT w4a16	22.6 tokens/s	not published

This corresponds to about 1 s of NPU time per 30 s of speech (roughly 30× faster than real time), so a one-hour lecture is transcribed in about two minutes. For comparison, the CPU fallback measured on a 13th-generation Intel Core i5 laptop runs the same model at 1.7–2.1× real time. The repository includes `scripts/benchmark.py`, which measures NPU versus CPU on the user's own Snapdragon laptop, and the app shows live timings in its Performance tab.

6. Deployment and accessibility

- **Two commands:** `setup.ps1` detects the chipset, installs dependencies and downloads the matching AI Hub model; `run.ps1` starts the app. No API keys or accounts.
- **Works offline** after setup, including fonts. Runs on any Windows PC through the CPU fallback.
- **Accessibility:** full keyboard operation, screen-reader landmarks and live regions, WCAG AA contrast, 44 px targets, light and dark themes, reduced-motion support, correct language tags for Indic scripts.

7. Verification status

Eleven unit tests pass, including tests of the decoder against the AI Hub input/output contract. The full pipeline, web API and user interface have been exercised end to end on the CPU path. The model I/O layout was checked against the published AI Hub assets. On-device NPU timings are produced by running `scripts/benchmark.py` on a Snapdragon laptop.

8. Technology

Python 3.11+, ONNX Runtime QNN, Qualcomm AI Hub models (Whisper-Small, Qwen3-4B-Instruct), Qualcomm GenieX, FastAPI, vanilla HTML/CSS/JavaScript. Licence: MIT. Source code, setup guide, design system (DESIGN.md) and benchmarks are in the GitHub repository submitted with this entry.